Development 1989 Supplement, 75–80 (1989)
Printed in Great Britain © The Company of Biologists Limited 1989

G-protein-linked signal transduction systems control development in *Dictyostelium*

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Summary

G-protein-linked cAMP receptors play an essential role in *Dictyostelium* development. The cAMP receptors are proposed to have seven transmembrane domains and a cytoplasmic *C*-terminal region. Overexpression of the receptor in cells, when the endogenous receptor is not present, results in a 10- to 50-fold increase in cAMP-binding sites. Antisense cell lines, which lack cAMP receptors, do not enter the developmental program. Ligand-induced phosphorylation is proposed to occur on serine and threonine residues in the receptor *C*-terminus. The kinetics of receptor phosphorylation and dephosphorylation correlate closely with the shift of receptor mobility and the adaptation of several cAMP-induced responses. Two α-subunits, G-α-1 and G-α-2, have been cloned and specific antisera developed against each. Both subunits are expressed as multiple RNAs with different developmental time courses. The mutant *Frigid A* has a functional defect in G-α-2 which prevents it from entering development. We propose that G-protein-linked receptor systems will be a major component in the development of many organisms.

Key words: G-protein, *Dictyostelium*, cAMP receptors, phosphorylation.

Role of cAMP signaling in morphogenesis and pattern formation

Dictyostelium is an elegant system for studying the role of cell–cell signaling in development. During vegetative growth, this organism exists as single amoebae that phagocytose bacteria. However, upon starvation, aggregation centers begin to secrete cAMP spontaneously at six minute intervals. Concentric or spiral waves of cAMP propagate from the centers and direct the formation of multicellular aggregates (Tomchik and Devreotes, 1981). As aggregates form 'slugs', cell-type-specific differentiation occurs and a pattern appears. Prestalk cells are found in the anterior region while prespore cells arise from the posterior of the cell mass (see chapter by Williams *et al.*). The prestalk cells become vacuolated to form a stalk, which holds a sporehead aloft in the fruiting body (Gerisch, 1987).

Aggregation is organized by closely coordinating cell–cell signaling and chemotaxis. As a wave of cAMP approaches, randomly oriented cells become polarized and move up the chemical gradient. As they move, the cells secrete additional cAMP which relays the signal outward. Cells continue to respond until the peak of the cAMP wave passes. They do not react to cAMP on the declining edge of the wave since they have become adapted. Hence secretion ceases and migration becomes random. During the interlude between cAMP stimuli, the system deadapts in preparation for the next wave. Approximately 30 waves are needed to complete aggregation (Devreotes, 1983).

Extracellular cAMP also plays a critical role in gene regulation. Expression of early genes such as the surface cAMP receptor, membrane-bound phosphodiesterase, and CSA (a cell adhesion protein) are induced by intermittent, but inhibited by persistent, application of cAMP (Figure 1). Transcripts of these early genes begin to accumulate within two hours and peak between six to eight hours of development (Mann and Firtel, 1987). In contrast, prestalk and prespore gene expression require continual, rather than intermittent, stimulation by cAMP. In addition, the morphogen DIF (Differentiation-Inducing Factor, see chapter by Kay *et al.*) is needed for prestalk gene expression in the tip where it also inhibits the induction of prespore genes (Schaap, 1983). Most prestalk genes are induced late in aggregation, while prespore gene expression begins somewhat later (Gerisch, 1987, see chapter by Williams *et al.*).

The molecular components involved in this signaling system include cAMP receptors, G-proteins, adenylate cyclase, and phosphodiesterase (Figure 2). These developmentally regulated proteins allow the establishment of the cAMP oscillator. Exogenous cAMP binds to surface membrane receptors which activate G-proteins and stimulate the formation of intracellular cAMP by adenylate cyclase. Each cell secretes cAMP which activates its

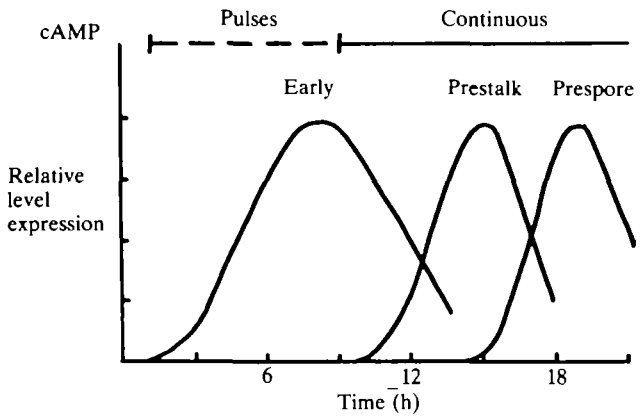

Fig. 1. Gene expression is controlled by cAMP stimulation. Many early genes are induced by intermittent stimulation by cAMP. In later development, prestalk gene markers are induced by persistent exposure to cAMP with prespore expression beginning a few hours later.

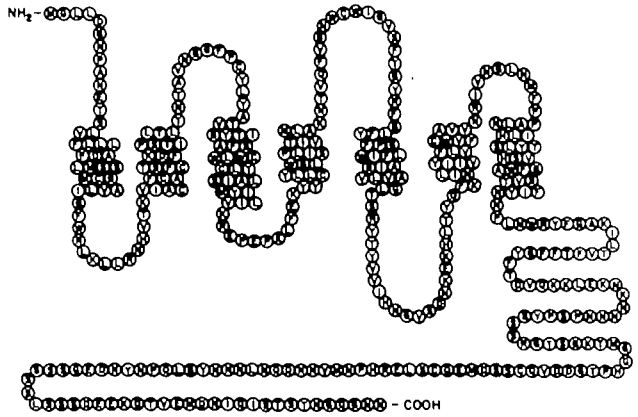

Fig. 3. The cAMP receptor consists of seven transmembrane domains with the *N*-terminus facing the extracellular and the *C*-terminus facing the cytoplasmic side of the membrane. Multiple stretches of serine residues in the *C*-terminal region are the proposed sites of phosphorylation.

own receptors as well as those of nearby cells. After several minutes of stimulation, ligand occupancy triggers desensitization (Janssens and Van Haastert, 1987). Removal of cAMP by extracellular and cell surface phosphodiesterase allows the system to resensitize and return to its prestimulus state (Gerisch, 1987).

Studies in our laboratory have focused on defining and characterizing the components involved in signal transduction. Recently, we have cloned surface cAMP receptors and proposed a model of their structure. We have also investigated ligand-induced phosphorylation. Two α- and one β-subunit of G-proteins have been cloned as well. Mutation and overexpression of these genes have helped to establish their essential roles in development.

Cell surface cAMP receptors

A structural model of the cAMP receptor was derived from its primary sequence (Figure 3). It is postulated to span the membrane seven times with the *N*-terminus facing the extracellular and *C*-terminal domain facing the cytoplasmic side of the membrane (Klein *et al.* 1988). The model is topologically similar to other

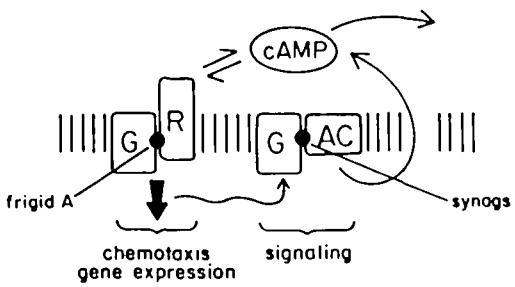

Fig. 2. The cAMP oscillator controls development. cAMP binds to cell surface receptors to activate adenylate cyclase and cause secretion of cAMP. *Frigid A*, which lacks a functional G α-subunit, is defective in chemotaxis and induction of gene expression. In *Synag 7*, receptors are unable to couple to adenylate cyclase.

G-protein linked receptors such as rhodopsin (Hargrave, 1986) and the β-adrenergic receptor (Dolhman *et al.* 1987). Although the *N*-terminus has a potential site for *N*-linked glycosylation, there is no biochemical evidence that the protein is glycosylated. The *C*-terminal region contains multiple stretches of serine residues which are the proposed sites of ligand-induced phosphorylation.

Multiple criteria demonstrate that this cDNA encodes the cAMP receptor. Antisera against the purified receptor identify several fusion proteins derived from the cDNA; RNA blot analysis, using the receptor cDNA as a probe, identifies a 2 kb message that is developmentally regulated in the same manner as the receptor protein; complementary receptor RNA specifically hybrid arrests *in vitro* translation of a 37 kD protein that is immunoprecipitated by receptor antiserum (Klein *et al.* 1988). Most conclusively, vegetative cells, which do not contain endogenous cAMP receptors, acquire cAMP binding sites when transformed with the cDNA. Transformed cells have seven to fifty times the amount of cAMP-binding compared to wild-type ($1-8\times10^5$ *vs* $1{\cdot}5\times10^4$ sites per cell, Johnson *et al.* in prep.).

To examine the function of the cAMP receptor in development, an antisense cell line that blocked production of receptor was created. A construct was designed to transcribe the receptor cDNA in the antisense orientation using a strong constitutive promoter. Antisense cell lines, which failed to express receptor RNA or protein, did not enter the developmental program. As shown in Figure 4, antisense cells remain as a smooth monolayer after eight hours of starvation while wild-type cells have developed into aggregates. By 36 hours, however, the antisense cells begin to show signs of weak aggregation (Sun *et al.* in prep.).

The cAMP receptor exists in two interchangeable forms designated R (40 kD) and D (43 kD). This

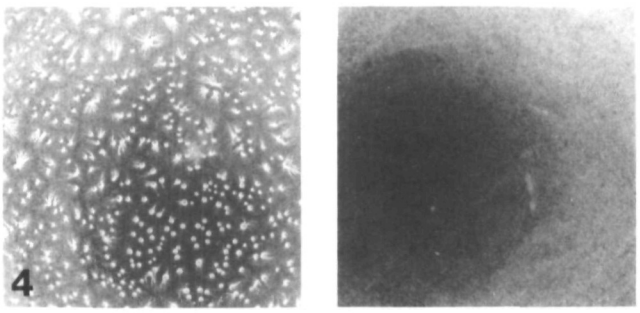

Fig. 4. Antisense cell lines, which lack cAMP receptors, do not enter the developmental program. The left panel shows wild-type cells forming aggregates after eight hours of starvation while antisense cells, shown in the right panel, are a smooth monolayer.

reversible shift of electrophoretic mobility is coupled to the spontaneous oscillations in cAMP synthesis in developing cells (Klein *et al.* 1985*a*). The R form is predominant in unstimulated cells and converts to the D form upon stimulation with cAMP (Devreotes and Sherring, 1985). Phosphorylation of both forms of receptor has been demonstrated *in vitro* and *in vivo*. Upon stimulation with cAMP, the phosphorylation increases by several-fold to a level of about four moles phosphate/mole receptor. It is likely that this phosphorylation underlies the shift in electrophoretic mobility (Klein *et al.* 1985*b*).

The kinetics of receptor phosphorylation and dephosphorylation correlate closely with the shift of receptor mobility and the adaptation of several cAMP-induced responses (Figures 5 and 6). Phosphorylation is detectable in five seconds, has a half time of 30 s, and reaches a steady state within five minutes. When cells are freed of cAMP, both the shift in receptor mobility and phosphorylation decay. Dephosphorylation is detectable in 30 s, has a half time of 2·5 min, and is complete within 15 min (Figure 6). The kinetics of receptor phosphorylation and dephosphorylation match those of adaptation and deadaptation of cAMP-stimulated activation of adenylate cyclase and myosin phosphorylation. The half-maximal responses of these events all occur with concentrations of 5 nM-cAMP (Vaughan and Devreotes, 1988). Receptor phosphorylation is probably essential in the adaptation response as demonstrated in other G-protein-linked receptor systems (Sibley *et al.* 1988). Isolation of a cAMP-receptor kinase mutant would directly test if receptor desensitization requires phosphorylation.

Predominately serine and some threonine residues of the cAMP receptor are phosphorylated (Klein *et al.* 1985*b*). Two serines are present on the third cytoplasmic loop and an additional 18 are found in the *C*-terminal domain (Figure 3). When membranes containing the phosphorylated receptor are trypsinized, a soluble fragment is released. This peptide has an apparent molecular weight of 19 kD and appears to contain all of the ligand-induced phosphorylation sites. Studies are now underway to isolate this fragment and

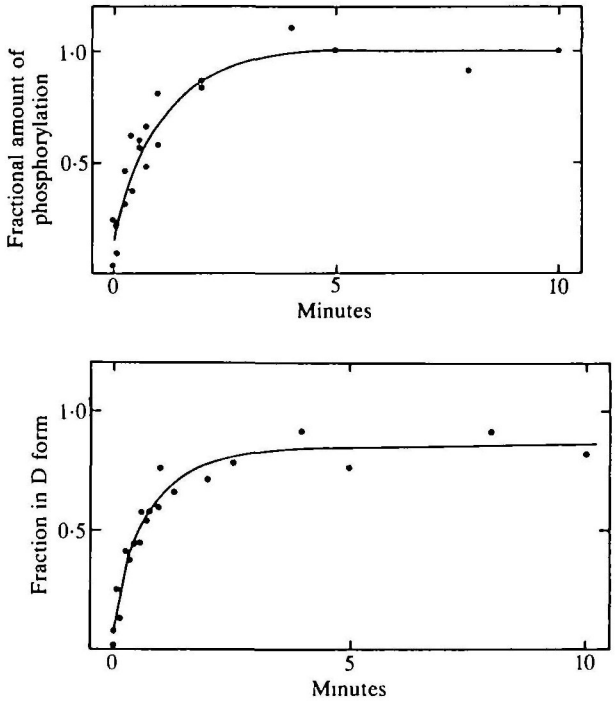

Fig. 5. The kinetics of phosphorylation (top panel) correlate closely with the receptor shift to the D form (bottom panel) during stimulation with cAMP. Both increase exponentially and reach a steady state by five minutes.

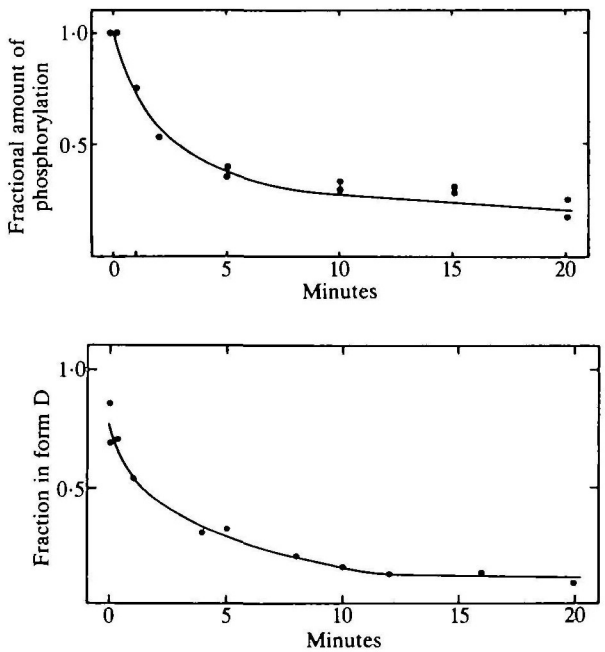

Fig. 6. The kinetics of dephosphorylation (top panel) and the shift to the R form of receptor (bottom panel) are similar after removal of cAMP.

map the sites of phosphorylation (Vaughan and Devreotes, in prep.).

Multiple G-proteins and signal transduction pathways

Two G-protein α-subunits, designated G-α-1 and G-α-2, have been isolated from cDNA libraries using oligonucleotide probes based on the highly conserved GTP-binding and GTPase sites of the α-subunits in mammals. The nucleotide sequences of both subunits encode polypeptides of approximately 41 kD. Both subunits are 45% identical to each other and to the mammalian α-subunits, G_i, G_o, and transducin, while only 35% identical to G_s and yeast GPA-1. When compared to the other α-subunits, some highly conserved regions are notable (Figure 7). Region A is almost 100% identical among all seven subunits and is the proposed GTPase site by analogy to c-Ha-ras and EF-Tu. Regions C, E and G are also highly conserved and are believed to form the GTP-binding site. A fifth domain, designated T, contains an almost completely conserved sequence of TCATDT and is near the *C*-terminus (Pupillo *et al.* 1989). This region may interact with receptor (Hamm *et al.* 1988).

Both G-α-1 and G-α-2 are expressed as multiple RNAs with different developmental time courses (Figure 8). The predominant 1·7 kb band of G-α-1 is present during vegetative growth, increases several-fold until the 'loose aggregate' stage and then decreases. In contrast, the major 2·7 kb RNA of G-α-2 is almost completely absent during vegetative growth and is induced during early development with peak expression in the aggregation state. The developmental regulation of G-α-2 is similiar to those genes involved in cell aggregation that require intermittent cAMP stimulation for induction.

Peptide antisera were developed that specifically recognize each α-subunit as well as a common domain of both proteins (Figure 9). When visualized by immunoblot, G-α-1 migrates as a 38 kD protein, while G-α-2 has an apparent molecular weight of 40 kD. The time course of protein expression is consistent with the transcription of the major mRNA for each subunit. G-α-1 is present in both vegetative growth and throughout development, while G-α-2 is expressed in early development, peaking around 6–8 hours in shaking cultures (Figure 9, Kumagai *et al.* 1989).

Characterization of the mutant *Frigid A* has helped to establish a function for G-α-2. *Frigid A* mutants bind cAMP, but appear to be blocked in subsequent steps of signal transduction (Coukell *et al.* 1983). Activation of adenylate cyclase and early gene expression fails to occur even when cells are stimulated with cAMP. However, adenylate cyclase can be activated by GTP *in vitro* (Kesbeke and Van Haastert, 1988; Kumagai *et al.* 1989). Both RNA blot and immunoblot analysis show

Fig. 7. All of the identified G α-subunits share conserved regions. Domain A contains the proposed GTPase site. Regions C, E, and G comprise the putative GTP-binding site. The T domain is thought to interact with receptors. (G1 – *Dictyostelium* G-α-1. G2 – *Dictyostelium* G-α-2. YE – Yeast GPA1. GO – Bovine G_o. GI – Rat G_i. TR – Bovine transducin. GS – Bovine G_s.

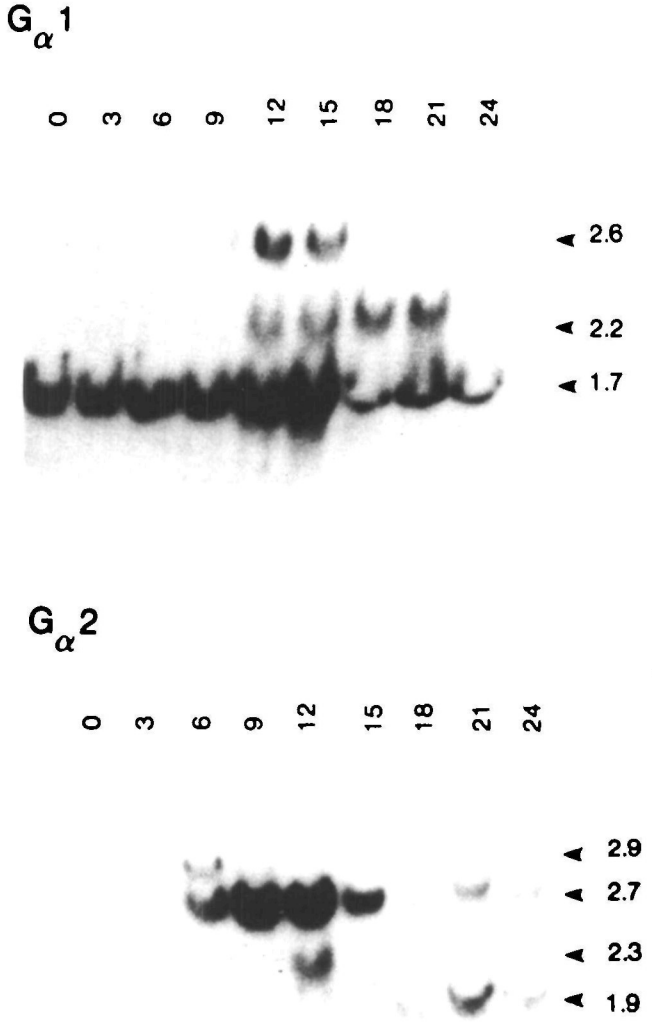

Fig. 8. The time course of G-α-1 and G-α-2 mRNA expression reveals multiple transcripts. The major 1·7 kb band of G-α-1 is present throughout development with peak expression between 12–15 h. The predominant 2·7 kb band of G-α-2 is developmentally regulated and expressed maximally at 12 h.

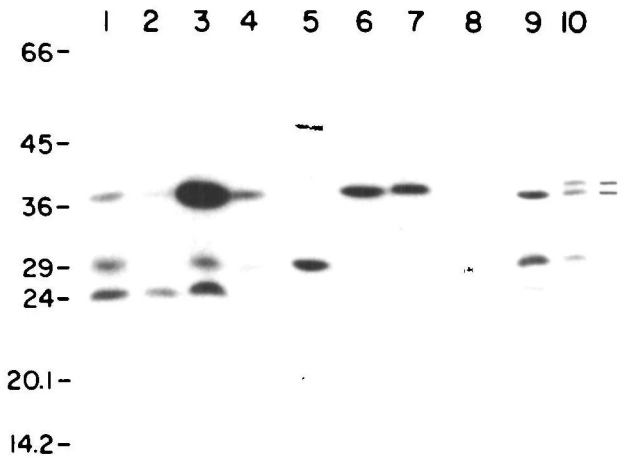

Fig. 9. Antibodies specifically recognize G-α-1 (lanes 1–4) and G-α-2 (lanes 5–8). In vegetative growth, G-α-1 is present (lane 1 & 9) but G-α-2 is absent (lane 5 & 9). Both α-subunits are present at six hours of development (lanes 2, 6, and 10). A cell line overexpressing G-α-1 shows a high increase in the staining of this protein (lane 3) while G-α-2 staining is normal at six hours of development (lane 7). *Frigid A* allele, HC 85, lacks G-α-2 (lane 8) when developed six hours. Both antibodies were mixed to stain G-α-1 and G-α-2 (lanes 9 & 10). The two bars show where G-α-2 (top bar) and G-α-1 (bottom bar) migrate.

most alleles to have a strong reduction or absence of G-α-2 mRNA and protein at all stages of cell development. Most conclusively, one *Frigid A* allele, HC 85, contains a deletion in the G-α-2 gene (Kumagai *et al.* 1987). Hence both biochemical and genetic evidence suggest that the *Frigid A* allele is the G-α-2 subunit. One *Frigid A* mutant, HC 112, however, has near normal levels of both mRNA and protein. The G-α-2 gene in this mutant was cloned by PCR (Polymerase Chain Reaction) and sequenced. It carries a point mutation within the coding sequence. Further characterization of HC 112 is in progress (Pitt and Devreotes, in prep.).

To investigate the function of the G-α-1 subunit, this protein was overexpressed. Both G-α-1 mRNA and protein levels are approximately 20-fold higher than wild-type as detected by RNA blot and immunoblots (Figure 9). Transformed cells grown on surfaces are large and multinucleate. When these cells are developed, most fail to aggregate but those that do, produce small and abnormal fruiting bodies. This phenotype suggests a role for G-α-1 in cytokinesis during growth (Kumagai *et al.* 1989).

A G-protein β-subunit has also been cloned using probes based on conserved sequences of *Drosophila* and human β-subunits. It is expressed as a constitutive 1·9 kb mRNA. The amino acid sequence has 63% identity to the human β-2-subunit. The first 50 residues share only 30% identity, with the remainder of the protein being 70% identical. The significance of these different domains is under investigation. Work is in progress to examine the function of the β-subunit by creating antisense and overexpression cell lines (Lilly and Devreotes, in prep.)

G-protein linked signaling pathways in development

Our studies of *Dictyostelium* show that G-protein-linked signal transduction pathways play a central role in the development of a multicellular organism. The cAMP receptors we have identified are essential to development; antisense cell lines do not enter the developmental program. The G-α-2 subunit is critical for cAMP chemotaxis and induction of gene expression; *Frigid A*, which lacks a functional G-α-2, is unable to differentiate. We propose that G-protein-

linked receptor systems will be a major component in the development of many organisms.

The oscillatory nature of cAMP-receptor activation and inactivation plays a critical role in cell–cell signaling, aggregation and gene expression in *Dictyostelium*. Positional signaling, *via* a cAMP oscillator, underlies cell aggregation in early development and establishes cell differentiation in later stages. Perhaps higher organisms such as *C. elegans*, *Drosophila*, or vertebrates use this type of cellular communication during embryogenesis. One could envision progenitor cells creating organization centers by establishing gradients or oscillations of a signaling molecule. These organization centers could influence local cell differentiation or act in concert to institute larger changes. Hence cAMP signaling may not just be limited to slime molds, but utilized in more complex organisms as well.

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